

Watt's Up, Sun

October 28, 2025

Leo Huang - ECE 110

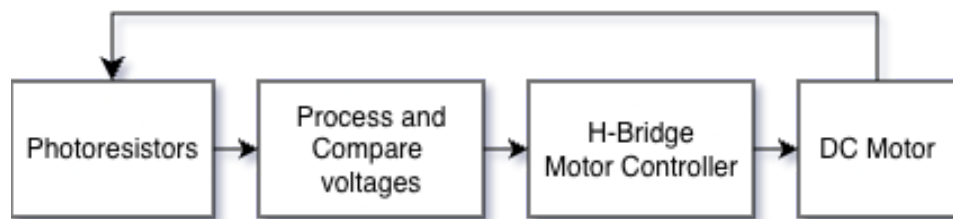
Andy Yang - ECE 110

Xuanyu Zhou - ECE 120

Abstract:

Solar panels don't always face the best direction depending on where you are and time of day as the sun moves through the sky. The goal is to be able to "track" the sun by seeing how the voltage from a photo sensor changes as time passes in order to always generate peak power with the solar panel.

Our goal is to use an array of 3 photoresistors to be able to accurately track the sun across the sky on one rotary axis, regardless of cloud cover. To do this, we will process the voltages across each photoresistor to decide whether or not to turn, and which direction to turn in order to maximize the light intensity on the center photoresistor.



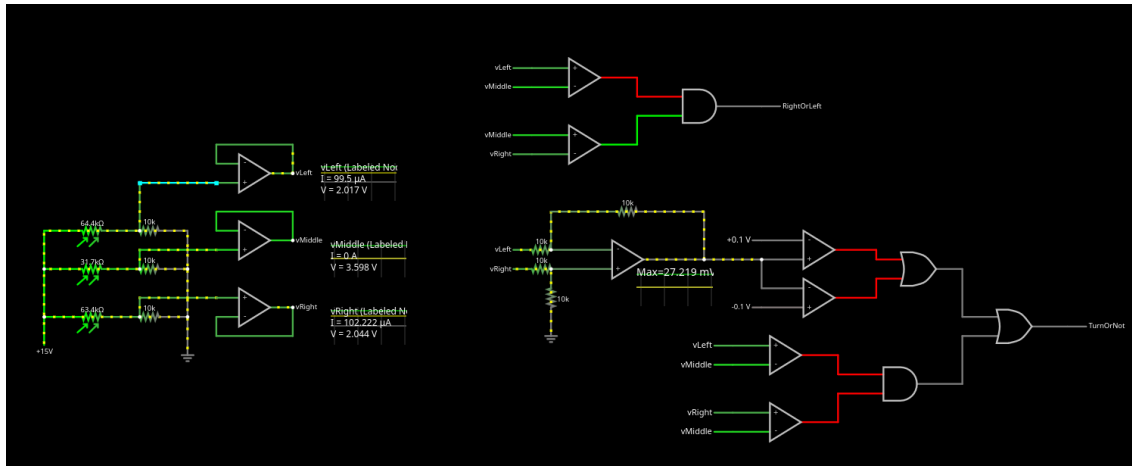
Timeline by Subproject/Subcircuit:

Schematic: (Responsible Individuals: All)

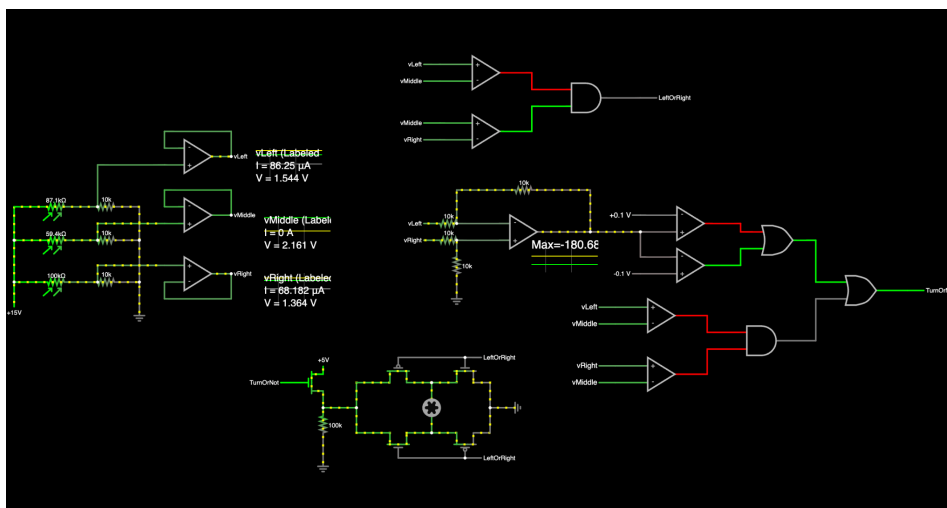
For the Schematics, it is in simulation, we just add something we came up with the project every week and update the schematics. It works as a guideline to our building process.

Week of 9/22

Preliminary Schematic



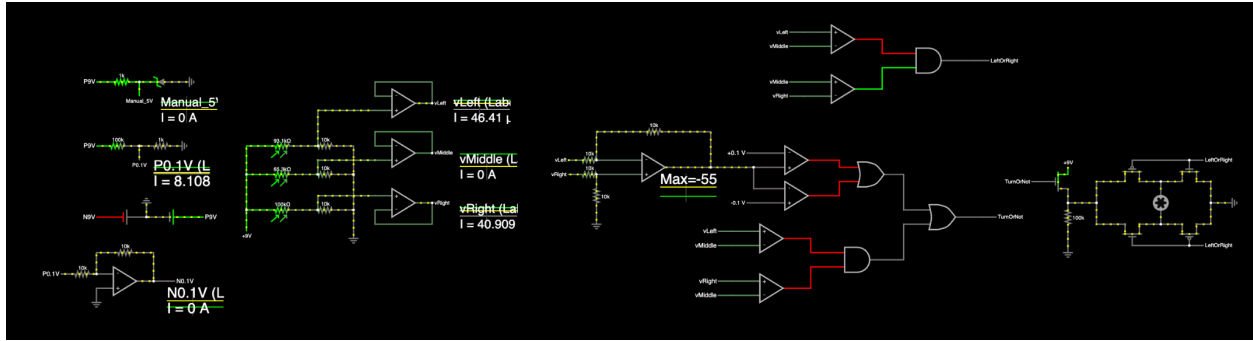
Week of 9/29



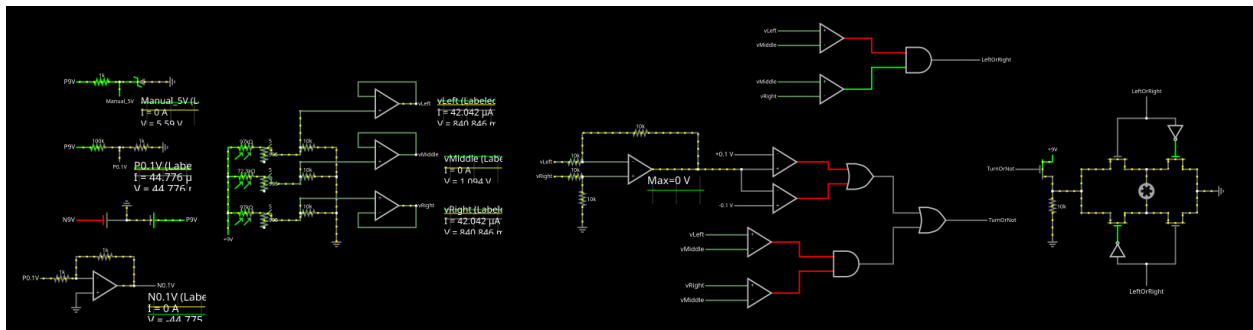
We added the H-bright to Schematics.

Week of 9/27

We add circuits to create voltages we need.



Week of 10/14

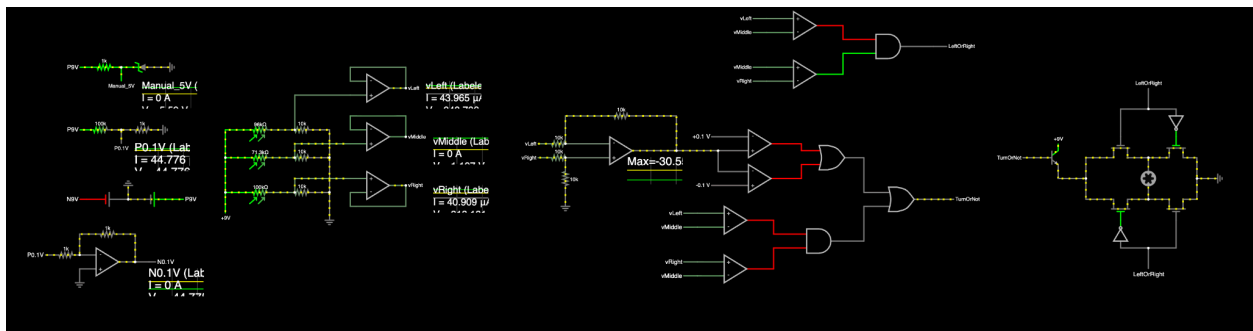


Add potentiometer to reduce individual difference between photoresistors.

Milestone: the schematics are completed in the week of 10/14 in a working state.

Week of 11/3

[Final Schematic](#)



Switched out switching MOSFET on H-Bridge for a BJT

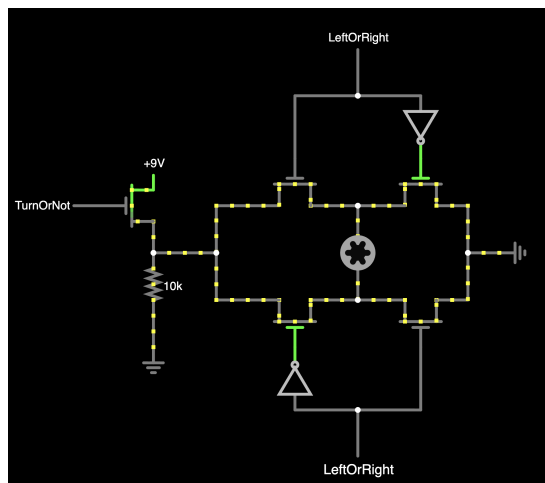
H-Bridge: (Responsible Individual: Leo Huang)

Week of 10/07

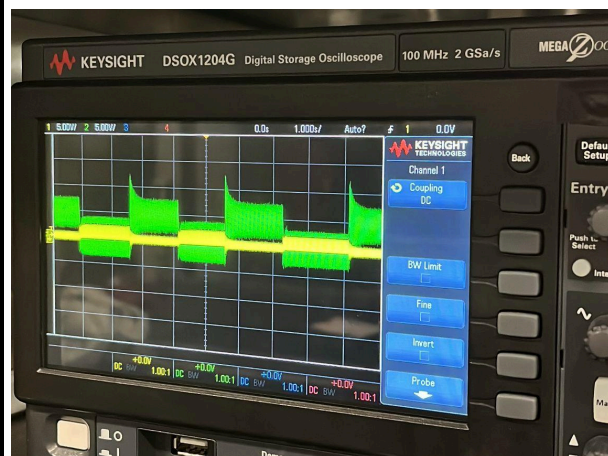
Completed schematic and majority of construction of H-Bridge Circuit, attempted to validate.

Challenges:

The oscilloscope modified the behavior of the circuit since I didn't account for the fact that the ground was shared on the oscilloscope, and connected two nodes accidentally that shouldn't have been connected. This is something NOT to do in the future.



H-Bridge schematic



Incorrect output from oscilloscope

Another challenge was that the voltage at the source of the input switching MOSFET was too low, which we solved later in the timeline.

Week of 10/14/25

Troubleshooting the H-Bridge circuit

Success: H-Bridge works separately from the switching MOSFET, isolating the problem to the switching MOSFET. This is how we found out the voltage at the source of the input was too low.

Challenges: When there was a load (the motor), the 5V power rail dropped to under 3V. In order to solve this problem, I added a voltage follower to keep a more consistent 5V power rail.

I was also still probing incorrectly, which I solved the next week.

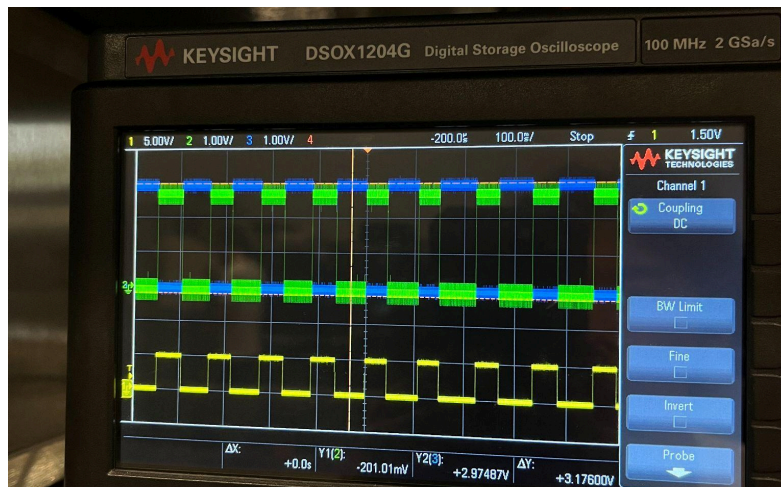
I also noticed periodic noise on the signals, but this was a byproduct of the inductive load of the motor, and didn't actually cause many problems.

Week of 10/20/25

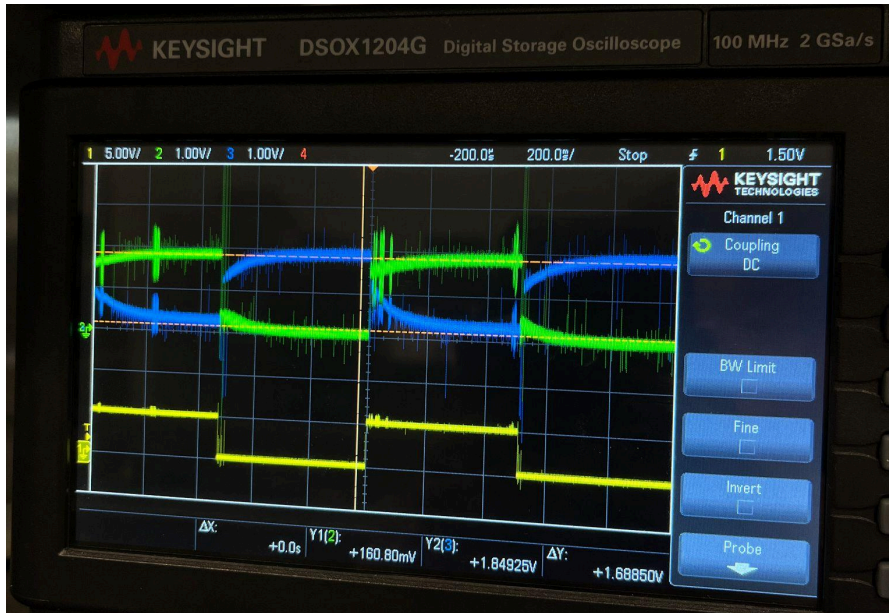
Validation + Operation + Successes:

Milestone: H-Bridge now works as intended!

The lower voltage at the source of the switching MOSFET due to its properties and function as a variable resistor also turned out to be a non-problem, since the voltage was still high enough to drive the motor. We verified the operation with both the behavior of the motor when operated as well as with oscilloscope readings, that matched what the output should be (The inductive load of the motor adds some curves to our square-wave but this is an unavoidable non-issue).



Without driving the motor: Yellow - Input, Green and Blue are probing opposite sides of the H-Bridge output

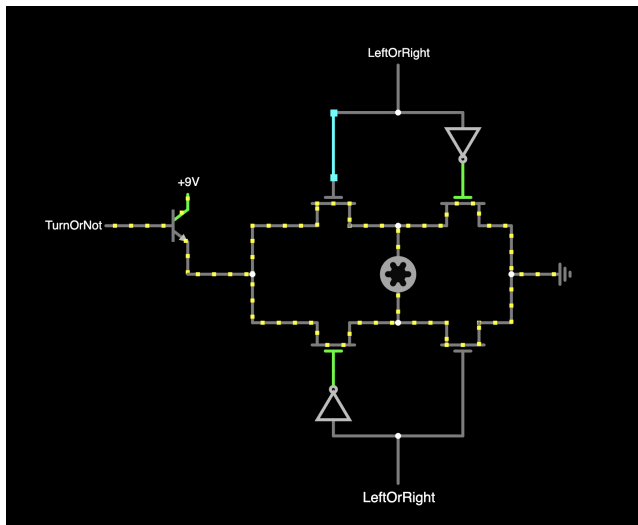


While driving the motor: Yellow - Input, Green and Blue are probing opposite sides of the H-Bridge output

Week of 11/3/25

Modification

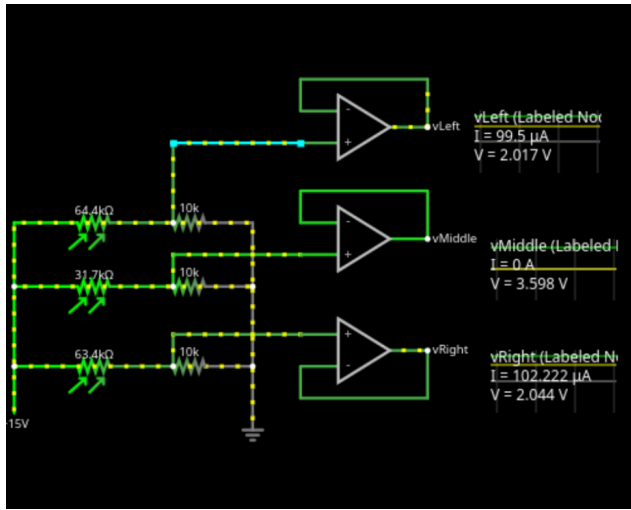
We replaced the MOSFET to BJT transistors to obtain more consistent output, as the MOSFET was acting like a resistor previously. The BJT gives a more definitive "high" behavior.



Input Circuit

Week of 9/22

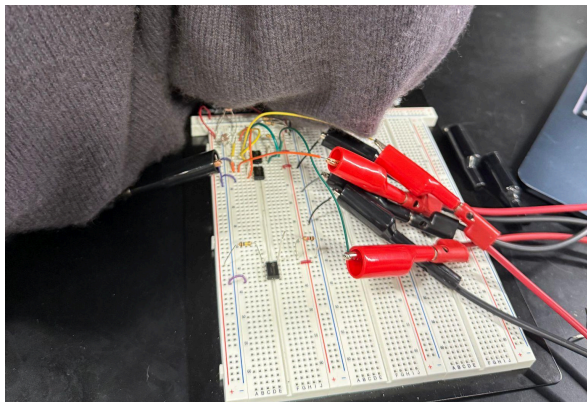
We came up with the first schematics of the all circuit



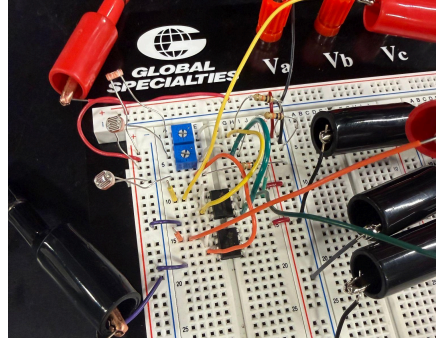
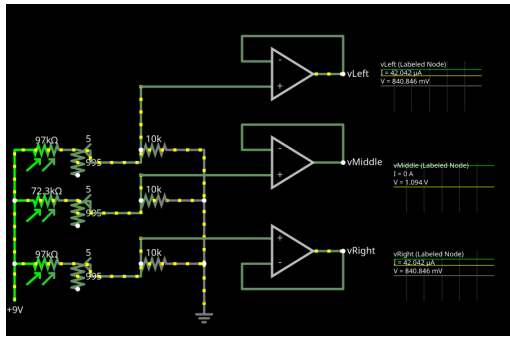
Challenges: We found that the subtractor changes the voltages of the input circuit. So we added a voltage follower built from op-amps to maintain the voltages and solve the problem.

Week of 10/14

Fully tested the input circuit: three different voltages on oscilloscope based on how much each photoresistor is covered from light.

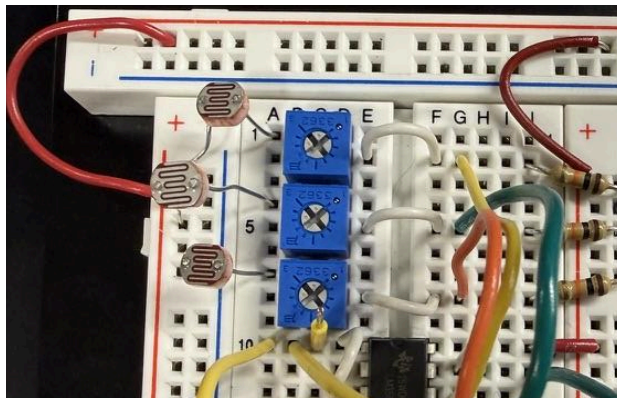


Add potentiometer in series with photoresistor to compensate for slight difference between each photoresistor.



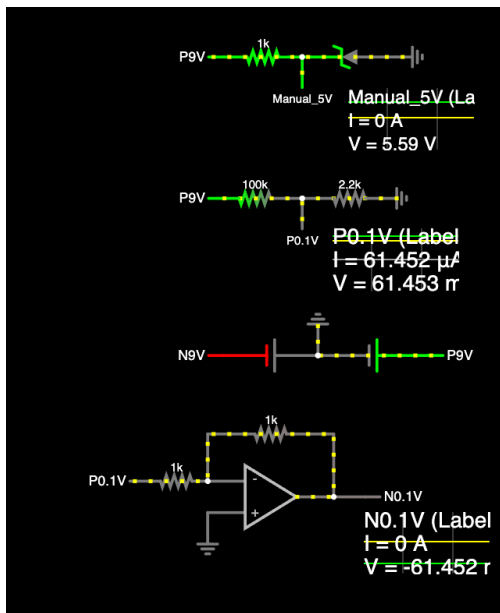
Week of 10/20

Add the last potentiometer which we left behind last week.



Milestone: The input circuit is complete !

Source Voltages



Our project is powered by a 9v battery, however, logic gates do not work under 9v. We need to find a way to step down the voltage and also our comparing logic needs a tolerance voltage.

Week of 10/07

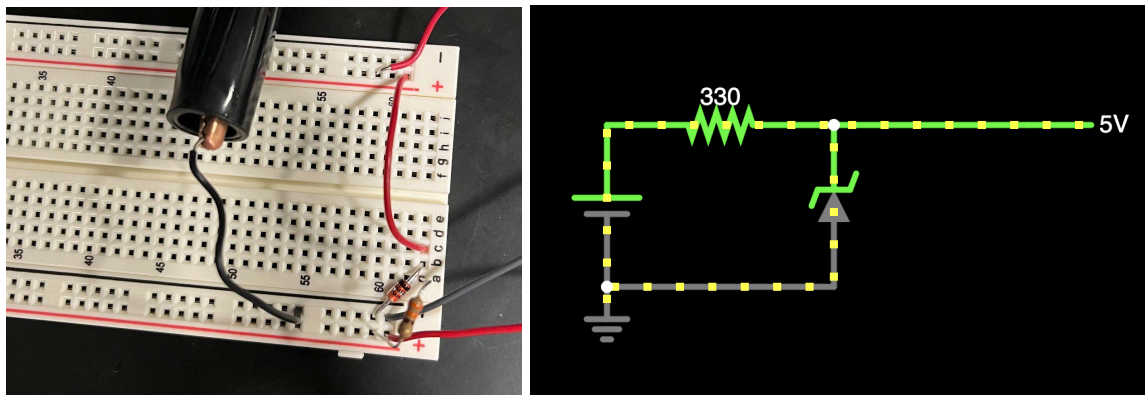
Come up with the schematics and tested 5v on the breadboard.

Schematics(Voltage Update)

For 9v and negative 9v we put two 9v batteries in series and connect the ground in between. +- 9v are mainly used for op-amps.

For 5v we use a zener diode, this is used for logic signals including AND gates and OR gates.

For +- 0.1v we created a voltage divider using 100k and 1k resistors. This voltage is used as a buffer between turning state and not turning state.



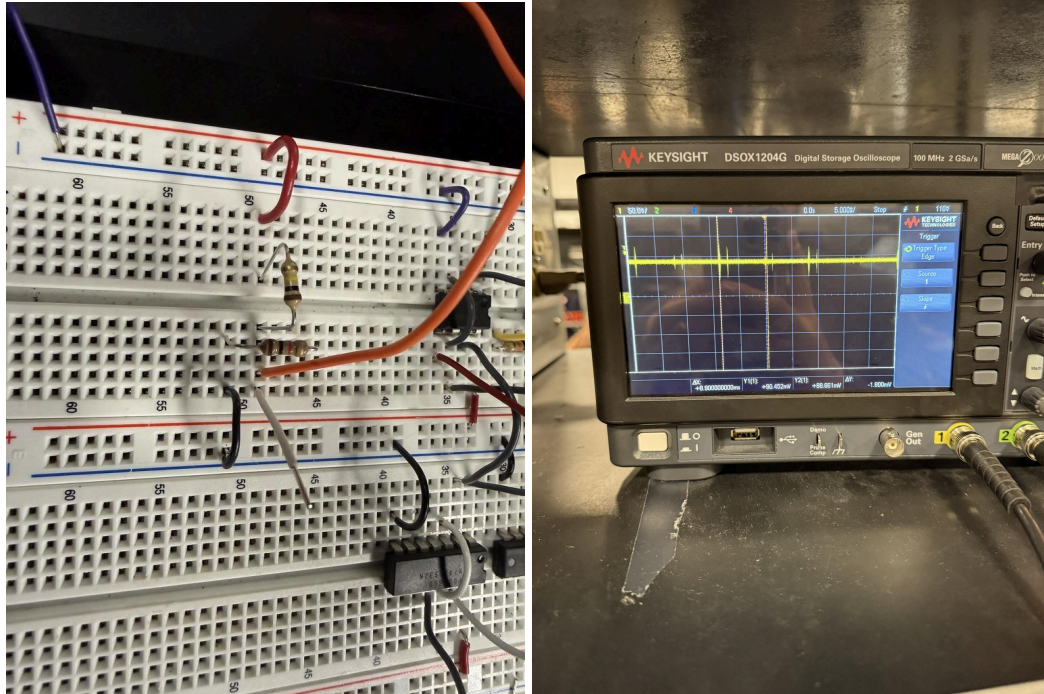
Using the digital multimeter on the bench, we validated that the output was approximately 5 volts and safe to use to power our ICs.



This matches the rating on the zener diode pretty closely, which had a breakdown voltage of 5.1V.

Week of 10/20

We implemented the voltage divider circuit to create 0.1 volt which we need to test comparator circuit later on.



Week of 10/27

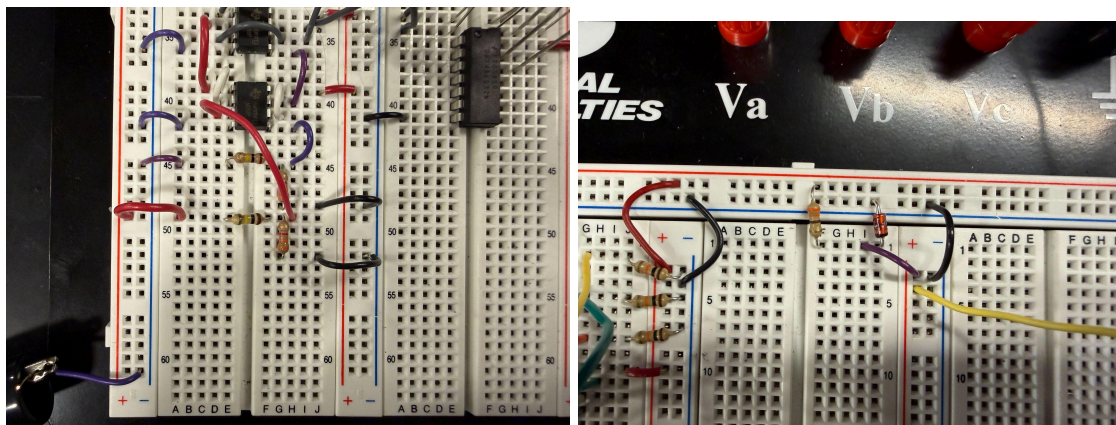
We implemented all the voltages we needed on the protoboard.

VERIFICATION, CHALLENGES, SUCCESS

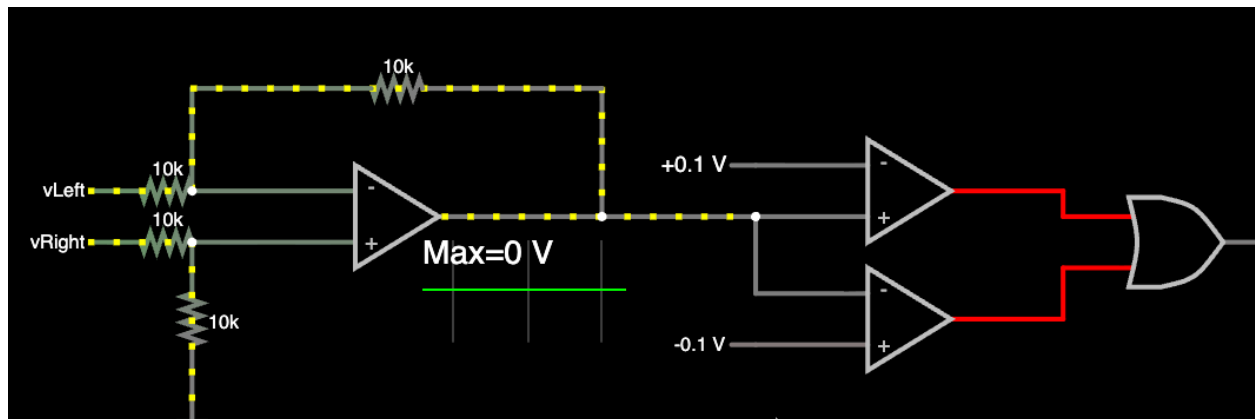
When validating the circuit, plus minus 0.1 is too small for our tolerance so we use 0.3 instead. Also using an inverting amplifier to get -0.3 would be too complex. So we used another potential divider instead followed by voltage follower to main voltage for further processing.

MILESTONE:

Before midterm, all the voltages are ready.



Subtractor followed by comparator and OR

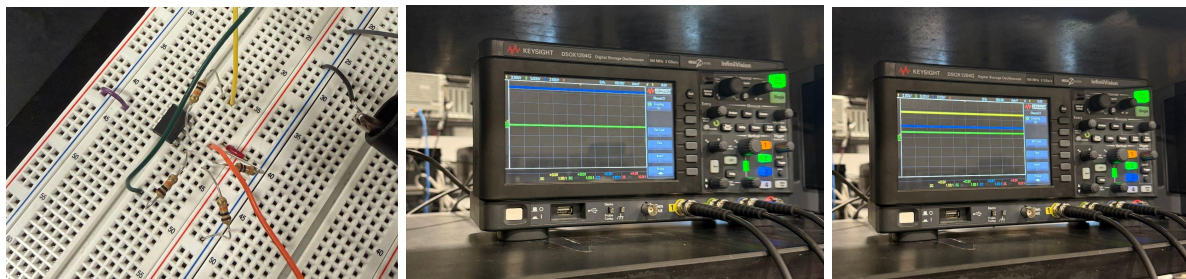


Week of 10/14

Challenges:

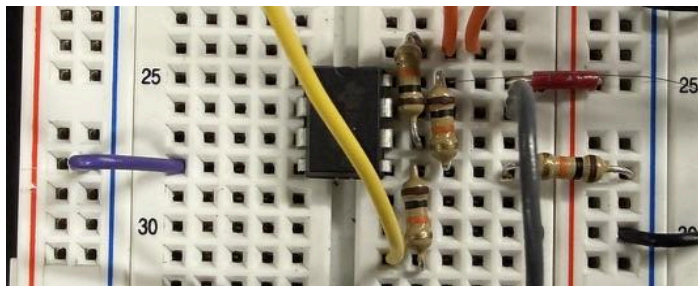
We started to build this part this week, but at first it is not working. So we isolated the subtractor and tested it. It works at the last minute.

Yellow and Blue signals are voltages from the input circuit. When they are the same, the green signal which is the difference is at ground level when red and blue signal are not the same, the green signal are slightly above the ground level at the difference between yellow and blue signal.

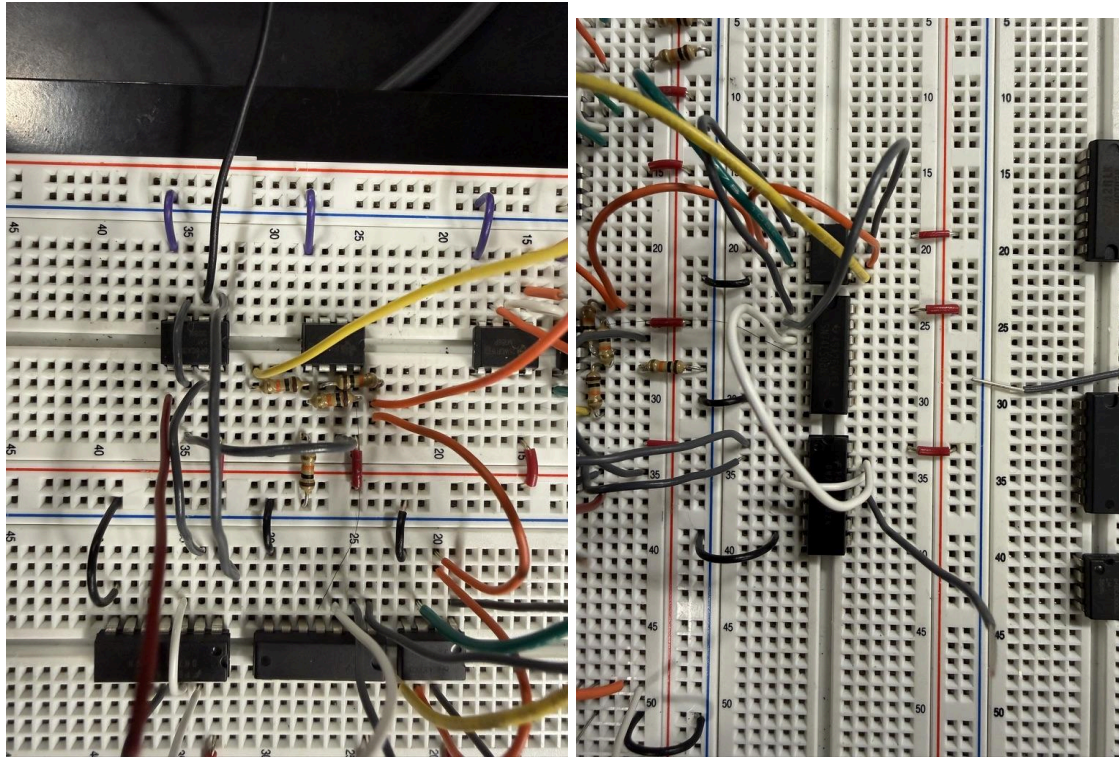


Week of 10/20

Last week we tested the subtractor at the last minute, this week we rewired it to make the wiring clean and clear.



This week we moved the comparator circuit from the previous breadboard where we isolated and tested on, to the main protoboard and connected it with the subtractor we tested before.



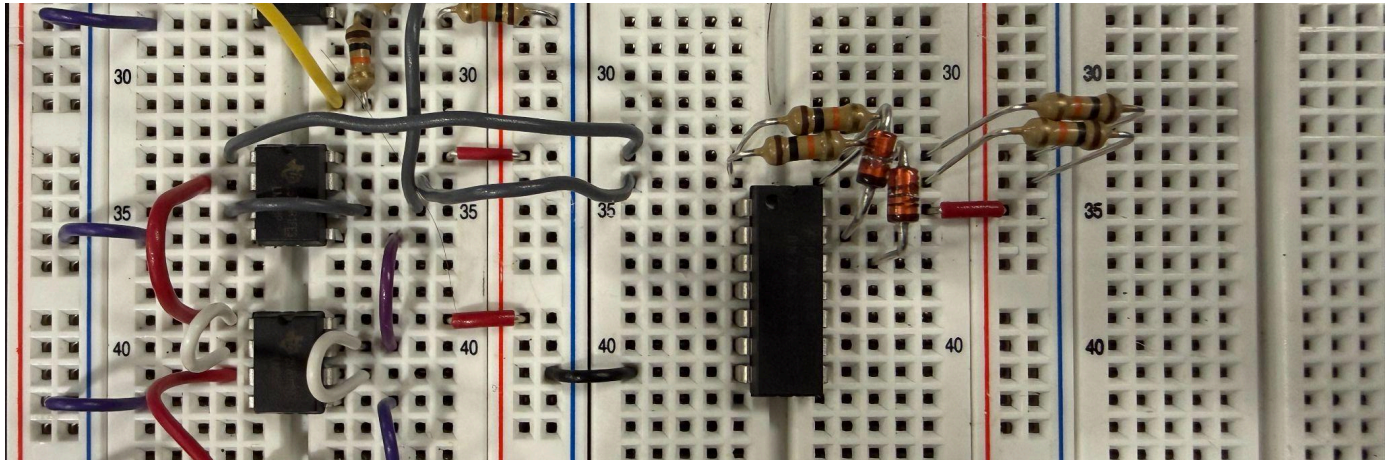
Week of 10/27

VERIFICATION, CHALLENGES, SUCCESS

We tested the circuit and it was not working. The problem is about the working voltage. Or gates work under 5v-0v for high and low signals. However, the output from the comparator is $\pm 9\text{v}$ for high and low signals. So we used a potential divider to take it to $\pm 4.5\text{v}$ and use a zener diode to make -4.5 to 0.

At first we tried a small diode which leaks current and does not work then we tried zener diode and it works.

MILESTONE

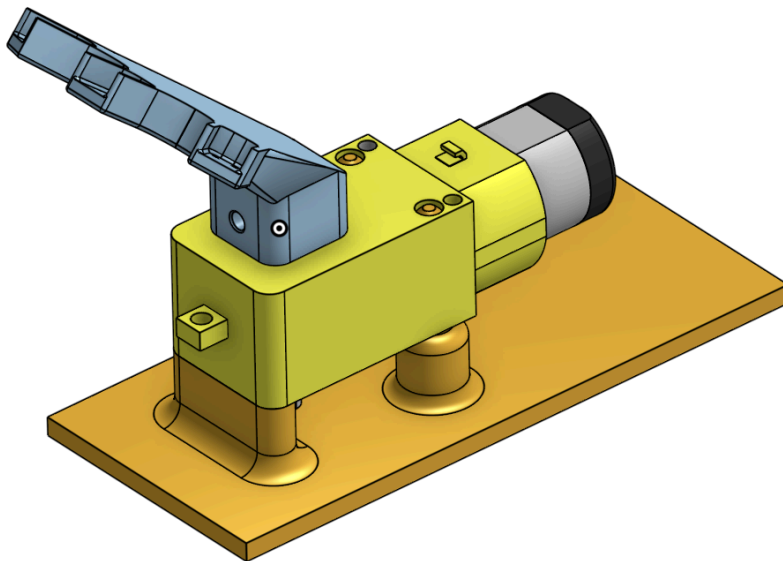


Looking ahead

Concerning that we need to step down every time we take output from opamp to logic gates, we might transplant everything to 5v in the future.

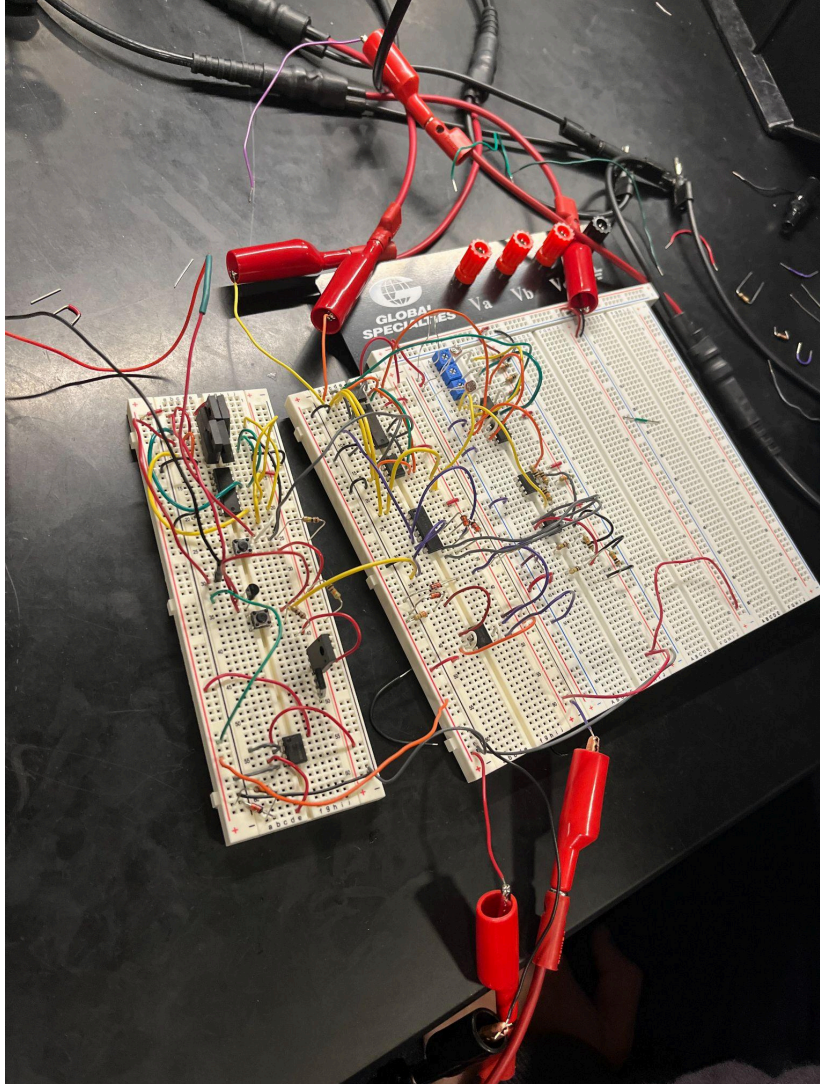
Week of 11/3

We finalized the design for the motor holder



Week of 11/10

This week marks a milestone for us as we combined all the components together with a variety of challenges that we needed to debug.



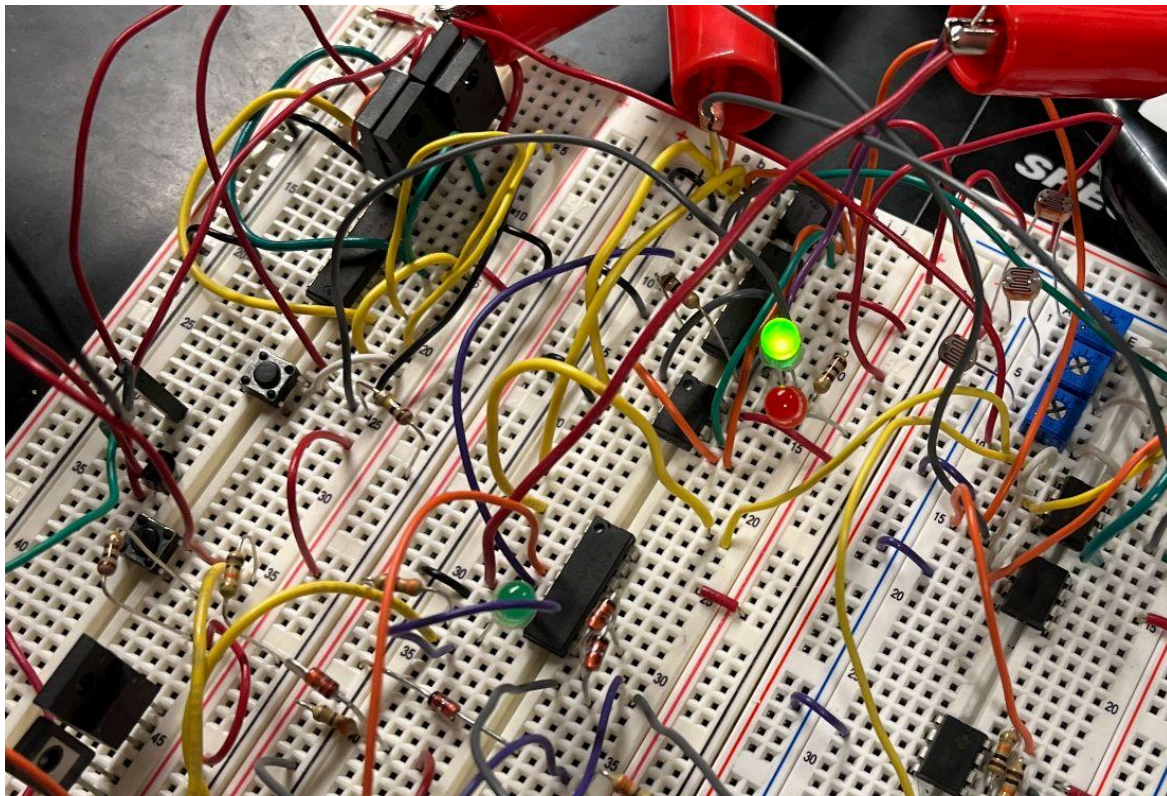
We made a couple of changes to the circuit regarding the power supply. Before we had +9 volts going into our logic and it was what was generally powering the whole circuit. Instead we transitioned to only needing to use +5 volts for the logic.

One major challenge is that although we all agreed on a formal color coding system to help navigate our circuitry, we were still struggling on which wire determined what the supposed output/input.

Week of 11/17

To make the debugging process easier, we added LED indicators to better read what is happening while our circuit is in operation. This lets us not need the oscilloscope to understand what is happening with the logic.

These LED indicators were following after the logic of the circuit to help us determine what the circuit wants to do even if we can't see the action itself.



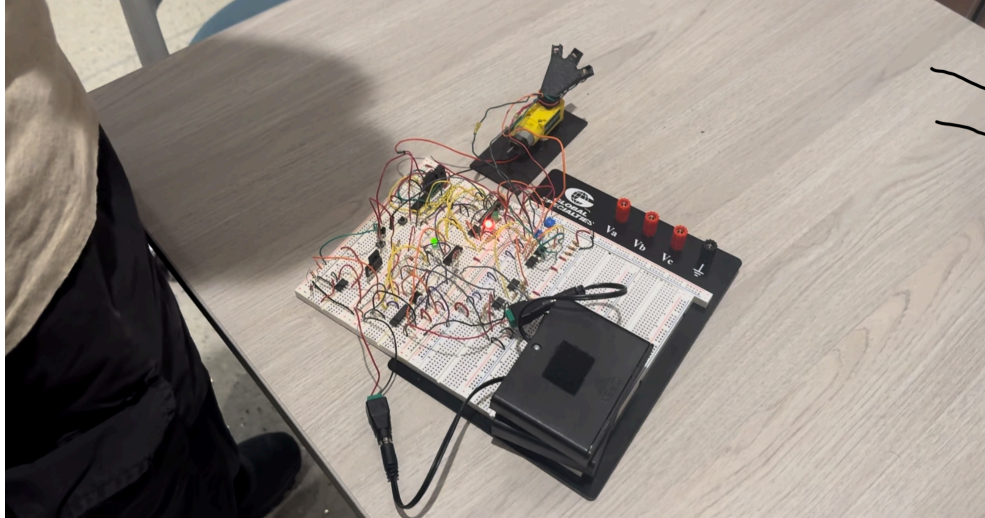
We had some issues with the output of the or-gates being inconsistent with what was expected. We probed voltages across the different logic gates and realized sometimes our “high” output was as low as 3.8V or even 3V. This was because the or gate was non-ideal and we had some loads on these logic gates. In order to combat this, we added schmitt trigger inverters on each of these to invert them to 0V and then back to 5V.

Week of 11/29

This week was a huge milestone for us as we assembled all our components together.

Week of 12/3

We began finalizing our testing and decided to use an artificial lightsource in order to validate our circuit, as it has been extremely cloudy recently.



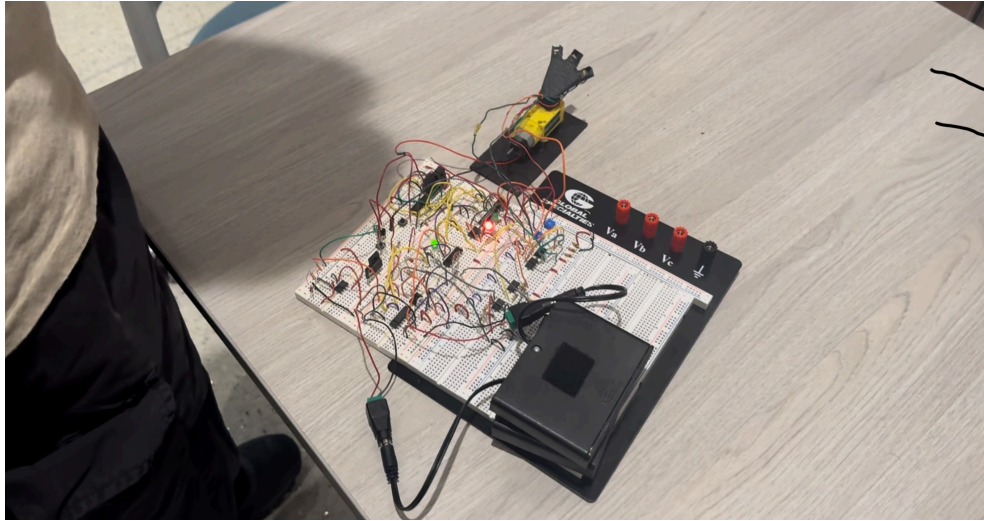
Challenges:

One challenge we noticed when wiring the whole thing was that the wiring that connects the photoresistor array to the circuit had some slack, but because of the characteristics of the solid core wire, the wiring acted as springs. As the motor assembly turned because of the logic, the wires would either be a source of compression or tension and force the assembly back to being centered, basically being a damped harmonic oscillator.

One way we tried was to increase the length of the wire; however, after giving longer wire the spring nature of the wires was still present, and still actively centered the photoresistor array.

We finally decided that we needed stranded wires instead of using solidcore as stranded wires allow more freedom of movement of the assembly.

Conclusion and Future:



The solar follower is able to follow a lightsource as we once wanted. We first started with an idea and drew a schematic in Falstad. We then created the components individually and then combined them altogether. We had a few hiccups and had to go to office hours in order to achieve our goal, but we were able to have a functioning prototype at the end.

While there are still a few discrepancies in our design, such as needing to switch out solidcore wires to stranded wires, the overall functionality of the solar follower remains the same.

While we were planning on doing a two axis assembly at the beginning of the year, we unfortunately did not have enough time to plan and prepare for another degree of movement. Regardless, we are proud of what we have accomplished.